PURPLE · NIGHT SHIFT LONDON · 2026-10-24 · BUILT AROUND MUBIT

# An AI race engineer redesigns the 2026 Red Bull RB22 to beat the real Silverstone pole

Same Red Bull Ford DM01 power unit · every car checked against FIA 2026 Section C · Mubit proposes the next change, a physics lap sim measures it, the measured result goes into the next call.

BEST LEGAL RB22 / DM01 CAR · MODEL PREDICTION ± BAND

**87.235** ± 0.154 s

95 % band 86.933–87.537 s · real 2026 pole ANT **88.111 s** (−0.876) · real HAD RB22 **88.746 s** (−1.511) · would be P1 across the band




RB22 baseline, built procedurally from the design's 23 parameters · "concept geometry built to the design's dimensions · FIA 2026 frame · not a team CAD model"

our car: physics lap sim · Silverstone · 2026 British GP qualifying conditions (docs/concept/CAMPAIGN.md §4.0) · real laps: real 2026 qualifying · OpenF1 / formula1.com

# What is new this weekend

FRESH-CODE RULE: ALL PROJECT CODE WRITTEN DURING THE EVENT

### PRE-EXISTING (NOT JUDGED AS NEW)

**Equation I**, our team's browser F1 game: `js/` `css/` `assets/` `vendor/` `tools/` `index.html`. Imported as one commit, `a711399`, Fri 2026-10-23 21:54 UTC, labelled "pre-existing".

PURPLE uses its 3D Silverstone, its car model and its race engine.

### NEW THIS WEEKEND (365 COMMITS SINCE FRI 22:47 BST)

- `concept/`: design layer, FIA checklist, physics lap sim, campaign, Mubit designer loop, 3D car, Silverstone replay
- `server/`: Hono API, the one Mubit client (`server/mubit.js`), engine dashboard routes
- `lab/`: the 5-knob research lab · `test/`: 29 suites, 326 tests · `docs/`
- **Game hooks, opt-in:** 7 game files, +633 / −48 lines: `?lab=1` (race result to the lab), `?concept=1` (concept replay, new `js/conceptrace.js`), aero inputs in `car.js` `ai.js` `carmodel.js` `race.js`, one script tag in `index.html`. Without the flags the game behaves as before.

# The problem

2026 IS A RULES RESET; A TEAM NEEDS TO KNOW WHAT TO TRY NEXT

- **The question.** Can a car that obeys FIA 2026 Section C (Issue 19, in force at Silverstone) and keeps its own power unit lap Silverstone faster than the real 2026 pole?
- **The baseline.** Red Bull RB22, Red Bull Ford DM01. Real qualifying: HAD **88.746 s** (P5); pole ANT (Mercedes) **88.111 s**.
- **Why it is hard.** 23 shape parameters plus materials, coupled through the rules (front-wing X+Z envelope, floor stall, mass and axle split, Straight Mode flaps) and through the lap (drag paid on every straight, grip earned in a few corners).
- **What an engineer needs** is the next change to try, with the evidence of the last ones, and a verdict that comes from a measurement, not from the model that proposed it.

**GROUND RULES WE SET OURSELVES**

- same engine: DM01 locked field by field (ICE 396.08 kW, MGU-K 350 kW, ES 4 MJ window)
- illegal car = no lap, no score
- the verdict is the simulator's, never the LLM's
- every number carries its source

real laps: real 2026 qualifying · OpenF1 / formula1.com · regulations: FIA 2026 F1 Technical Regulations Section C, Issue 19 · PU values: docs/concept/CAMPAIGN.md §4.0

# How it works

REAL DATA IN, A LEGAL CAR OUT, EVERY STEP MEASURED

**1 · REAL DATA**
OpenF1 qualifying telemetry (22 laps at Silverstone, 9 circuits) · FIA Section C · formula1.com times

**2 · THE CAR**
RB22 estimated from public telemetry; 23 shape parameters, 23 materials; DM01 locked

**3 · LEGAL?**
94-row FIA 2026 checklist in `regs.js` + same-engine and material guards

**4 · LAP**
physics lap sim, Silverstone, 2026 British GP qualifying conditions: lap, sectors, per-corner Δ

**5 · KEPT?**
legal and faster, computed from the sim's numbers

↑ proposals come from two engines ↓ results go back

**MUBIT DESIGNER TASK (THE JUDGED LOOP)**
Claude Haiku 4.5 through Mubit, frozen prompt v1. Each call receives the current best car + the last 3 tested changes with measured lap, sector and segment deltas, kept or not, FIA articles. Returns 1–3 parameter changes. One Mubit session per run, one request id per result.

**SPEED RUN (NO LLM)**
CMA-ES on the physics sim with a surrogate pre-screen, on worker threads, 37 laps/s on a loaded 4-CPU box. Finds the fastest legal car.

**AUTONOMOUS MODE**
joins the two: Mubit sets the direction, a legal-only local search of up to 200 physics laps exploits it.

↓

**ENGINE DASHBOARD**
`/concept/lab/` → Engine: speed run · autonomous Mubit · garage · RB22 car sheet

**3D CAR**
built from the parameters at true size; hot swap of any part or material

**SILVERSTONE REPLAY**
the sim's own lap on the game's 3D track, ghosts: RB22 and ANT's real pole lap

docs/concept/LOOP.md · DASHBOARD.md · SIM.md · FIA_CHECKLIST.md · MATERIALS.md · RACE.md · laps/s measured in DASHBOARD.md §2 (6,740 laps in 184 s)

# The Mubit loop, one real call

TRAINED RUN SEED 11 · SESSION LOOP-01M5PGPKGSPFT7DTX7DZ1GQRRE · ITERATION 3

```
POST api.mubit.ai/anthropic/v1/messages · claude-haiku-4-5 · max_tokens 250 · temperature 0
system: designer prompt v1 (frozen, 994 chars) · metadata.user_id: the run id = one session
user message, 1,592 chars ≈ 398 tokens (line breaks added, … = cut here):
{"best":{"lap_s":88.656,"gap_s":0.545,"kg":770,"front":0.45,"params":{…23 by part…}},
 "tested":[
  {"change":"survival_cell.bulkhead_gap_mm 380->360; floor.ride_height_mm 35->33.5;
             rear_wing.sm_delta_deg 15->20",
   "d_lap_s":-0.096,"ds":[-0.02,-0.104,0.028],"kept":true},
  {"change":"rear_wing.aoa_deg 20->20.5; ballast.x_frac 0.05->0.04",
   "d_lap_s":0.041,"ds":[0.014,0.077,-0.05],"kept":false,"by":"fallback",
   "gain":["Hangar Str -0.053","Wellington Str -0.023"],
   "loss":["Copse-Maggotts +0.041","Maggotts-Becketts-Chapel +0.032"]}],
 "refused":[{"change":"front_wing.sm_delta_deg 4->6; sidepods.inlet_cm2 650->620;
             rear_wing.endplate_top_mm 860->840","why":"illegal C3.10.10(s)(i)"}],
 "near":["floor.ride_height_mm 33.5 (min 33.5 PKG-FLOOR-STALL)",
         "front_wing.span_mm 1300 (max 1300 C3.10.1(a))", …7 more]}
```

| CALL MODE | WHAT MUBIT KEEPS (PER ITS DOCS, LOOP.MD §3) |
|---|---|
| untrained (`x-middleware-record: metadata`) | a record without request or reply; never a kept example |
| **trained** (this call) | request + reply in the **designer task**, session = run id; may become a kept example (first adapter trains at 25) |

**← MUBIT ANSWERS · MREQ_01A16D0B7587…**

`rear_wing.aoa_deg 20→19.5` · `sidepods.inlet_cm2 650→630` · `ballast.x_frac 0.05→0.06`

"Reduce rear wing AoA for lower drag on straights; moderate sidepod inlet …"

**= THE PHYSICS LAP SIM MEASURES**

**88.656 → 88.638 s (−0.018)** · sectors −0.012 / +0.058 / −0.065 · gained Hangar Str −0.094 · legal · **KEPT**

**→ FED TO THE NEXT CALL**

This result is `tested[]` in iteration 4's request `mreq_01a16d0b80e1…` (lineage `loop.fed_to`). Trained calls store it in Mubit with the next proposal.

docs/concept/loop-runs/trained-seed11/lineage.jsonl row 3 (`loop.scenario`, `loop.mubit_request_id`, `loop.fed_to`) · laps: physics lap sim · Silverstone · 2026 British GP qualifying conditions (sim v2 at the time of this run)

# Mubit in numbers

EVERY CALL HAS A REQUEST ID; EVERY RESULT KNOWS WHICH LATER CALL CARRIED IT

| MUBIT DESIGNER RUNS | RUNS (SESSIONS) | ITERATIONS | MUBIT CALLS | TRAINED | LLM PROPOSAL USED | KEPT (PROPOSED BY LLM) | BEST LAP · Δ BASELINE |
|---|---|---|---|---|---|---|---|
| live untrained, prompt tuning | 3 | 13 | 13 | 0 | 7 of 13 | 3 (3) | 88.667 · -0.085 |
| trained, prompt v1 frozen (sim v2) | 4 | 60 | 60 | 60 | 52 of 60 | 9 (9) | 88.612 · -0.140 |
| trained, big run (sim v3) | 12 | 300 | 271 | 271 | 199 of 271 | 66 (39) | 88.292 · -0.471 |

- **331 trained designer calls, 331 unique request ids, 16 sessions**; 315 results fed into a later call (the last of each run has no successor).
- Refused answers (63 illegal, 9 duplicates in the big run) are replaced by one seeded fallback step and marked `by: "fallback"` in the next input; 29 iterations made no call (designer paused after 3 failures).
- Project budget: 99 research + 331 designer = **430 of 2,500** trained calls. Median latency 2.3 s, ≈ 1,030 tokens in, 191 out.
- Mubit loop alone, 25 calls a run: best 88.292 s, still +0.181 s to pole. The 87.235 s car comes from the speed run.

**MUBIT CONSOLE · DESIGNER TASK (OWNER READS)**

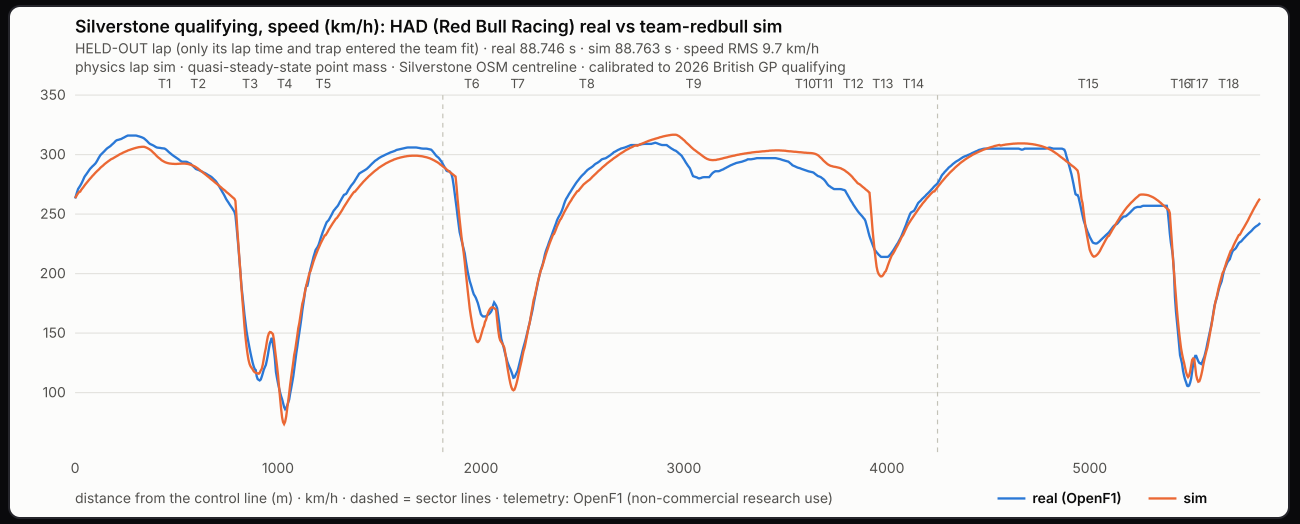
- kept examples: **FILL designer-kept**
- adapter status / test result: **FILL designer-adapter**
- Try models, same scenarios, legal-and-faster proposals: provider **FILL n/N** · adapter **FILL n/N**

docs/concept/loop-runs/*/{calls,lineage,summary} as of 2026-10-24 09:53 UTC · laps: physics lap sim · Silverstone · 2026 British GP qualifying conditions · baselines 88.752 s (sim v2) / 88.763 s (sim v3) · budget: EVAL_GUARDRAILS §4

# The physics

SPEEDS COME OUT OF THE PHYSICS; TELEMETRY ESTIMATES THE UNKNOWNS AND CHECKS THE RESULT

- **Physics lap sim v3**: quasi-steady point mass with axle load transfer, power-law tyre load sensitivity, ride-height aero and floor stall, Straight / Corner Mode only in the FIA zones, the C5.12 power-reduction rules, wind, density, gradient. ~21 ms a lap. No per-corner factors.
- **Estimated on one lap** (ANT's pole trace), checked on the other 10 teams' laps: held-out speed RMS **10.1 km/h** (target < 7: **missed**), sector bias +0.172 / +0.015 / −0.192 s. RB22 baseline 88.763 s vs HAD's real 88.746 s.
- **FIA checklist**: 94 rows against Section C Issue 19 (45 checked, 9 when a material is set, 13 partial, 4 not checked, 23 fixed / not applicable); an adversarial oracle on ~1,450 designs agrees rule by rule.
- **Materials**: 23 materials with their C15 reading; survival cell must be CC-UHS. **Coupling**: all 23 shape parameters reach the simulated car with a plausible sign.
- **Identification**: 11 cars over 9 circuits, 194 OpenF1 laps, bootstrap mean ± std.




HAD's real 2026 qualifying lap (OpenF1) vs the sim's Red Bull car, speed over distance, RMS 9.7 km/h

physics lap sim · Silverstone · 2026 British GP qualifying conditions (docs/concept/SIM.md §0, §7) · FIA_CHECKLIST.md summary · MATERIALS.md · COUPLING.md · IDENTIFY.md · telemetry: OpenF1, an independent community project

# The result

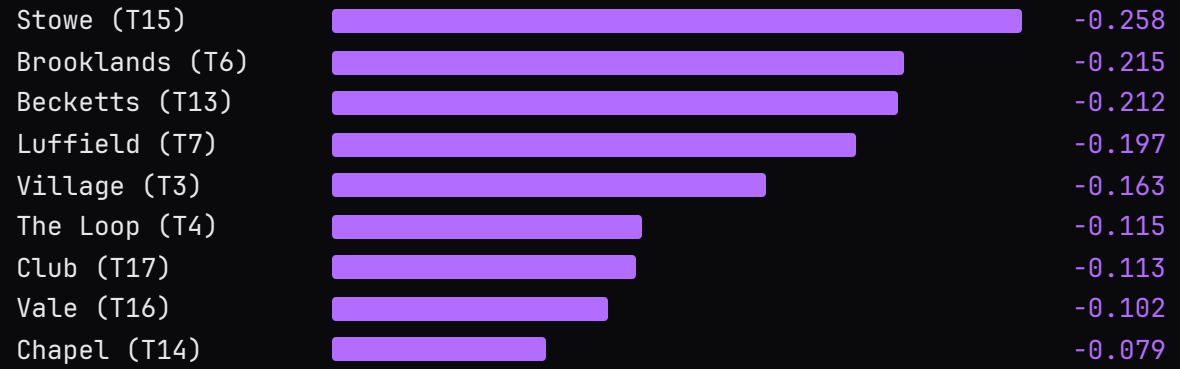
BEST LEGAL RB22 / DM01 CAR · MODEL PREDICTION ± BAND

| | LAP (S) | Δ POLE 88.111 | Δ HAD 88.746 | GRID |
|---|---|---|---|---|
| RB22 baseline (sim) | 88.763 | +0.652 | +0.017 | P6 |
| best legal car | 87.235 ± 0.154 | -0.876 | -1.511 | P1 (95 %: P1–P1) |
| aero map gains halved | 87.893 | -0.218 | -0.853 | |

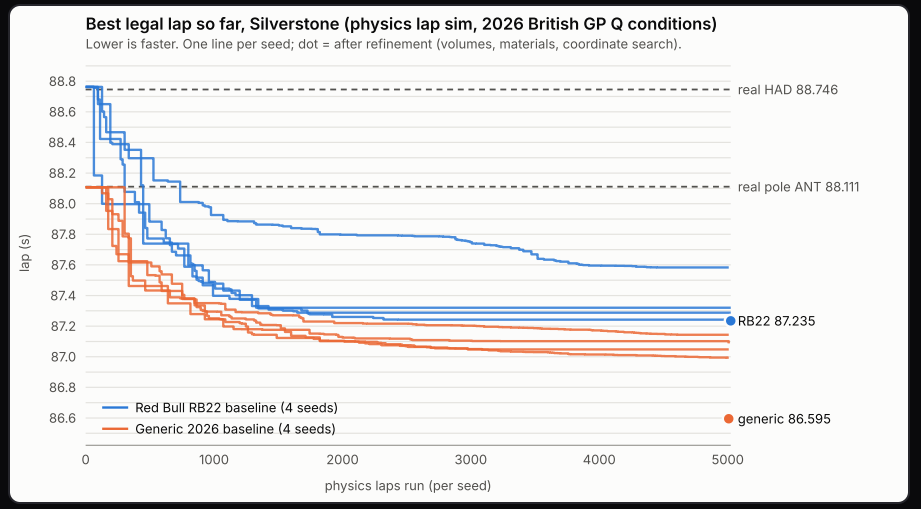
95 % band **86.933–87.537 s**: the whole band is below the real pole. 4 seeds × 5,000 physics laps + refinement.

- **PU identical** to the homologated DM01 (ICE 396.08 kW, MGU-K 350 kW); energy 7.633 MJ deployed, 3.657 of 6.5 MJ recharged.
- **At FIA limits**: floor width 1540, rear-wing span 1150, endplate 880, front-wing span 1350, bulkhead gap 360 mm. Floor throat 66.5 mm (roof 75).
- **Worth most**: Straight Mode flap angles −0.381 s, sidepods −0.276 s.

## WHERE THE 1.528 S COMES FROM (PER CORNER, BEST − BASELINE)



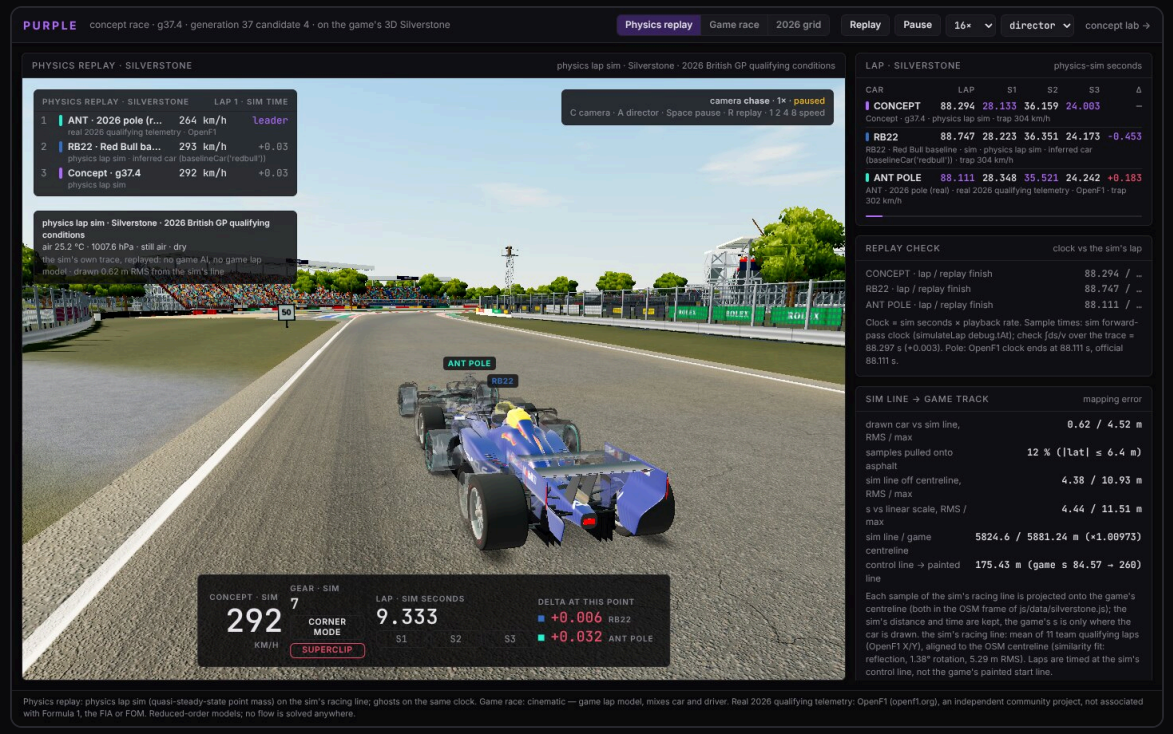

Stowe minimum 214.2 → 245.9 km/h, Becketts T13 197.6 → 221.7 km/h. The fast straights lose a little (T9→T10 +0.038 s): more downforce, more drag.




physics lap sim · Silverstone · 2026 British GP qualifying conditions · docs/concept/CAMPAIGN.md §4.0 (sim v3 + FIA geometry fixes, 673.9 s run) · real laps: real 2026 qualifying · OpenF1 / formula1.com

# See it: garage, 3D car, Silverstone

HOT SWAP A PART, WATCH THE LAP MOVE, THEN WATCH THE CAR LAP SILVERSTONE




**Silverstone replay**: the physics sim's own speed trace on the game's 3D Silverstone (drawn 0.62 m RMS from the sim's line); ghosts RB22 (sim) and ANT's real pole lap (OpenF1). Shown: loop candidate g37.4, 88.294 s.




**3D car**: built from the design at true size (meshes within 1 mm of the parameters on 11 test designs), baseline ghost and from → to tags; hot swap rebuilds one part in 10–23 ms. Not a team CAD model.

- **Garage** (Engine → Garage): change any parameter or permitted material; FIA verdict, lap and per-corner deltas come back in ~200 ms. "Watch on Silverstone" replays the current best car; the replay finish equals the sim lap.
- The game-race and 2026-grid modes are labelled **"cinematic · game lap model, mixes car and driver"**; they are not results.

garage and 3D: docs/concept/DASHBOARD.md §4, §5 · VIEW.md (dimension test, hot swap) · replay: docs/concept/RACE.md (mapping error) · laps: physics lap sim · Silverstone · 2026 British GP qualifying conditions · pole lap: real 2026 qualifying · OpenF1

# First proof that Mubit learns from a human: the 5-knob lab

BUILT FRIDAY NIGHT · THE FIRST MUBIT TASK · `/lab/`

- **Task "Research engineer of F1"**: Claude Haiku 4.5 through Mubit, frozen prompt v1 (sha 6c5be1a3). Input: baseline and candidate setup (5 knobs), the evaluator's deltas, active pit-wall rules, last 3 experiments. Output: verdict + the single next experiment.
- **99 trained calls**: 57 on Friday (43 kept examples, Training v1), 1 demo experiment, 41 on Saturday carrying the engineer's rule; 61 untrained (smoke, exam).
- **Rule-conditioned learning.** The engineer overrules a decision; the rule *"Weigh straight-line drag against corner gain; judge on full-lap delta."* enters every later task input; the provider's answers change; those kept examples train the next adapter version. In the v2 collection the provider's verdicts agreed with the evaluator on 38 of 39.
- **Exam** (20 scenarios, truth from the evaluator): provider verdicts 20/20, next experiment 10/20. Try-models sheet (8 hardest): provider 8/8 verdicts, 2/8 next step (7/7, 1/7 without exam-rw8, the demo item).
- **Race check**, rear wing +8°: predicted +0.436, measured +0.518 game s (seed 42, n = 1).

**MUBIT CONSOLE · RESEARCH TASK (OWNER READS)**

- kept examples after v2: `FILL research-kept`
- adapter v1 / v2 test result: `FILL research-adapters`
- Try models, 8-item sheet, next step v1 · v2: `FILL n/8 (n/7)`

Concept-lab lesson taken from here: the result of each decision must travel in the next input, because Mubit learns only from what it records.

reduced-order lap model · calibrated to 2026 pole laps · aero map v1 · game seconds (80.800 s = real pole 88.111 s) · call counts: docs/ledger-snapshot/2026-10-24_02-37-26Z-0530-windcap-exam/calls.jsonl · race: measured in race, seed 42

# Limits and honesty

READ BEFORE QUOTING A NUMBER

- **The sim misses its own accuracy target**: held-out speed RMS 10.1 km/h against < 7; S1 +0.17 s and S3 −0.19 s bias; Copse → Becketts runs 10–19 km/h fast, Brooklands 19.5 km/h slow. Corner-mode downforce sits outside its sourced prior (5.32 vs 4.6 m²; widened, reason in SIM.md §4.2).
- **The band does not cover the concept aero map**: its component shares and slopes are estimates. With every aero gain halved the car laps 87.893 s; it is a reduced-order model, not a flow solution, and no part of it was measured on a real car.
- **Part masses** are labelled estimates (tub, floor, tyre masses are not public); the Nominal Tyre Mass is not published.
- **Mubit has no outcome-feedback API.** The result travels in the next call's input; Mubit stores it only on trained calls, and adapters imitate kept examples. We cannot tell Mubit "this proposal was good".
- **The LLM is not the oracle**: it predicted a faster lap for 250 of its 251 used trained proposals; the sim measured 48 of them faster (those 48 were kept). 72 of 271 trained answers in the big run were refused (illegal or duplicate). The simulator decides.
- **The Mubit loop alone** reached 88.292 s (+0.181 to pole); the 87.235 s car is the speed run's.
- **Equation I is pre-existing**; game-race modes are cinematic. "Model prediction ± band", never "we beat F1".

SIM.md §0 §7 · CAMPAIGN.md §2 §5 · MASS.md · PRIORS.md · LOOP.md §4 (Mubit docs) · designer `expected_hit` in docs/concept/loop-runs/trained-seed*/lineage.jsonl, as of 09:53 UTC

# What's next

MORE OF THE SAME LOOP

- **Train and test the designer adapter**: the 331 trained calls in 16 sessions are the first examples; compare provider and adapter on held-out sessions as n/N, then run the autonomous mode on the adapter.
- **Outcome-filtered replay** (needs Mubit's OK): re-send only the calls whose proposal was kept, so the stored examples lean toward proposals that worked.
- **Close the sim's gap**: transient corner dynamics and the energy policy after Copse; then the held-out RMS target (< 7 km/h).
- **More circuits**: telemetry and track models for 9 circuits are already in the repo; a car has to be fast across a season, not one lap.
- **Any team's car**: the same pipeline takes any of the 11 identified 2026 cars as the baseline.

Run it: `npm ci` · `server/.env` with `MUBIT_API_KEY` · `npm run dev` → `127.0.0.1:8788/concept/lab/` (Engine) · `/concept/race/` · `/lab/` · deployed: `FILL deployed URL` · `github.com/rayhankhilji/purple`

counts: docs/concept/loop-runs/ as of 2026-10-24 09:53 UTC · telemetry: OpenF1 (9 circuits, docs/concept/TRACK.md) · identification: docs/concept/IDENTIFY.md